

Exp 4 Design of Low Pass Filter using Windowing Techniques

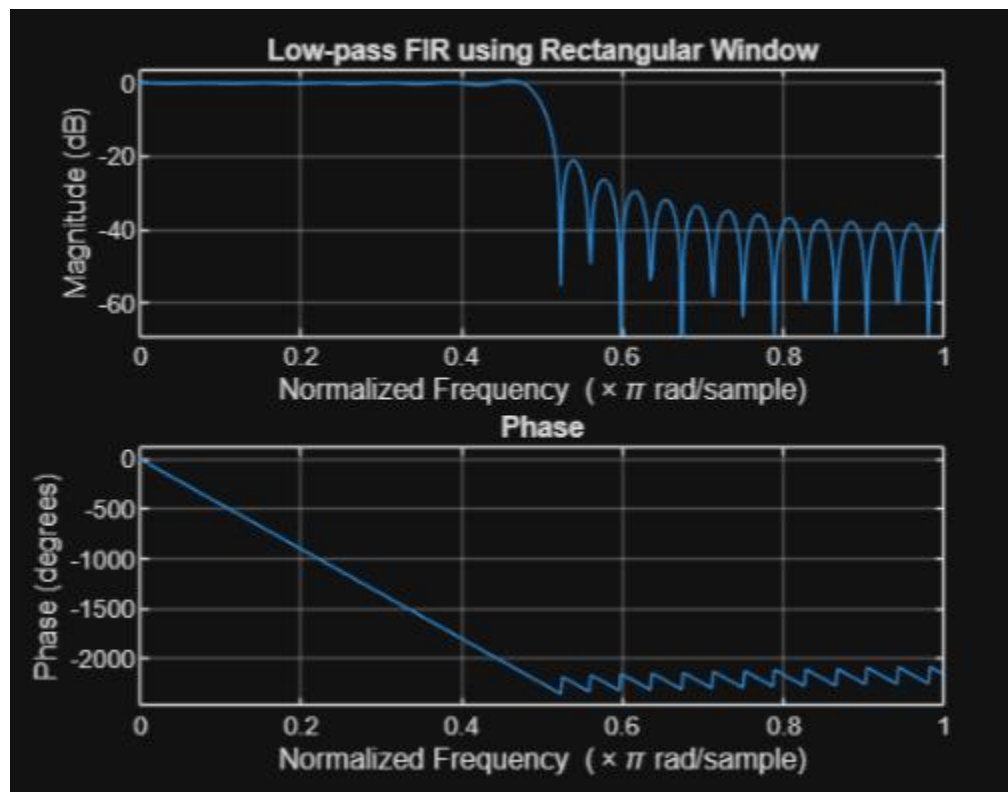
4.1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect');
%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');
%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hann(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
%% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

--- Rectangular Window Coefficients ---

```
0.0126
0
-0.0137
0
0.0150
0
-0.0166
0
0.0185
0
-0.0210
0
0.0242
0
-0.0286
0
0.0349
0
-0.0449
0
0.0629
0
-0.1048
0
0.3145
0.4940
0.3145
0
-0.1048
0
0.0629
0
-0.0449
0
0.0349
0
-0.0286
0
0.0242
0
-0.0210
```



```
0
0.0185
0
-0.0166
0
0.0150
0
-0.0137
0
0.0126
```

Result

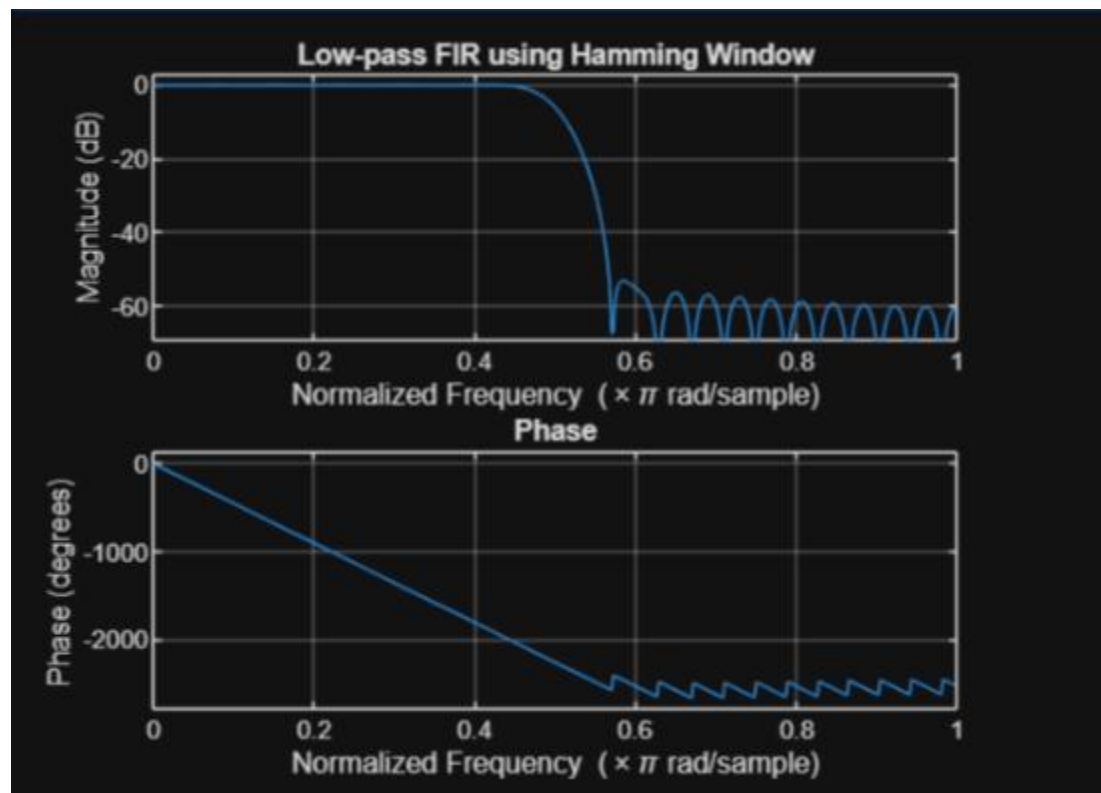
1. An FIR Low-Pass Filter was successfully designed using the Rectangular Window technique for the specified cutoff frequency, sampling frequency, and filter order.
2. The FIR filter coefficients were successfully generated using MATLAB and the designed filter was implemented.
3. The magnitude response and phase response of the designed FIR Low-Pass Filter were obtained and analyzed.

4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique in MATLAB

```
1  clc;
2  clear;
3  close all;
4  %% --- Input Parameters ---
5  fp = 1000; % Passband cutoff (Hz)
6  fs = 4000; % Sampling frequency (Hz)
7  N = 50; % Filter order (even number preferred)
8  wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9  %% --- FIR Low Pass using Hamming Window ---
10
11  b_hamm = fir1(N, wc, 'low', hamming(N+1));
12  disp('--- Hamming Window Coefficients ---');
13  disp(b_hamm);
14  %% --- Plot Frequency Responses ---
15  figure;
16  freqz(b_hamm,1);
17  title('Low-pass FIR using Hamming Window');
```

--- Hamming Window Coefficients ---

0.0010
0
-0.0013
0
0.0021
0
-0.0034
0
0.0055
0
-0.0084
0
0.0125
0
-0.0181
0
0.0260
0
-0.0379
0
0.0580
0
-0.1026
0
0.3168
0.4995
0.3168
0
-0.1026
0
0.0580
0
-0.0379
0
0.0260
0
-0.0181
0
0.0125
0
-0.0084
0
0.0055
0
-0.0034



```
0
0.0021
0
-0.0013
0
0.0010
```

Result

1. The FIR Low-Pass Filter was successfully designed using the Hamming Window technique for the specified filter parameters.
2. The FIR filter coefficients were generated and implemented successfully in MATLAB.
3. The magnitude and phase responses of the designed filter were obtained and analyzed.
4. The filter exhibited the desired low-pass characteristics with improved stopband attenuation compared to the Rectangular Window method.

4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB

```

1  clc;
2  clear;
3  close all;
4  %% --- Input Parameters ---
5  fp = 1000; % Passband cutoff (Hz)
6  fs = 4000; % Sampling frequency (Hz)
7  N = 50; % Filter order (even number preferred)
8  wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9  %% --- FIR Low Pass using Hanning Window ---
10 b_hann = fir1(N, wc, 'low', hanning(N+1));
11 disp('--- Hanning Window Coefficients ---');
12 disp(b_hann');
13 %% --- Plot Frequency Responses ---
14 figure;
15 freqz(b_hann,1);
16 title('Low-pass FIR using Hanning Window');

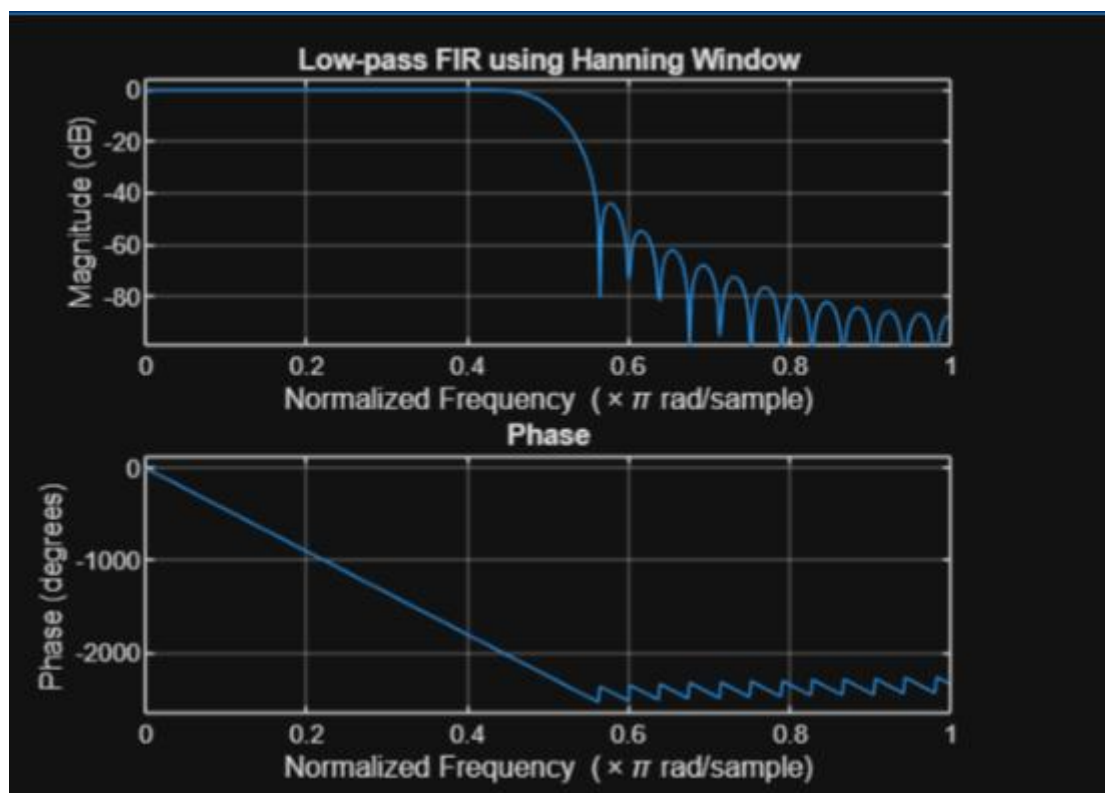
```

--- Hanning Window Coefficients ---

```

0.0000
0
-0.0004
0
0.0013
0
-0.0028
0
0.0050
0
-0.0081
0
0.0122
0
-0.0179
0
0.0259
0
-0.0378
0
0.0580
0
-0.1027
0
0.3172
0.5000
0.3172
0
-0.1027
0
0.0580
0
-0.0378
0
0.0259
0
-0.0179
0
0.0122
0
-0.0081
0

```



```
0
0.0050
0
-0.0028
0
0.0013
0
-0.0004
0
0.0000
```

Result

1. The FIR Low-Pass Filter was successfully designed using the Hanning Window technique for the specified filter parameters.
2. The FIR filter coefficients were generated and implemented successfully in MATLAB.
3. The magnitude and phase responses of the designed filter were obtained and analyzed.
4. The filter exhibited the desired low-pass characteristics with reduced sidelobes and improved stopband attenuation compared to the Rectangular Window method.